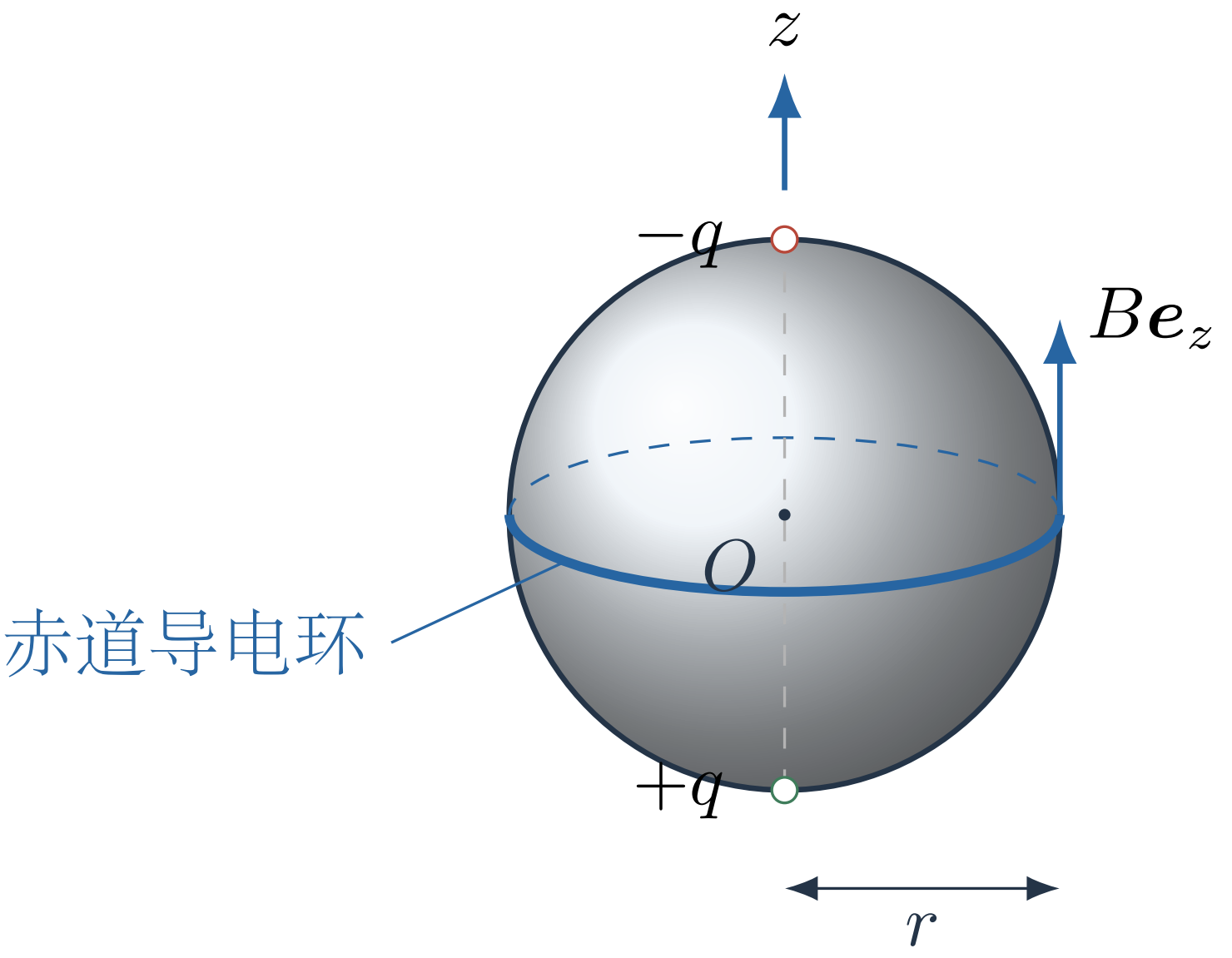
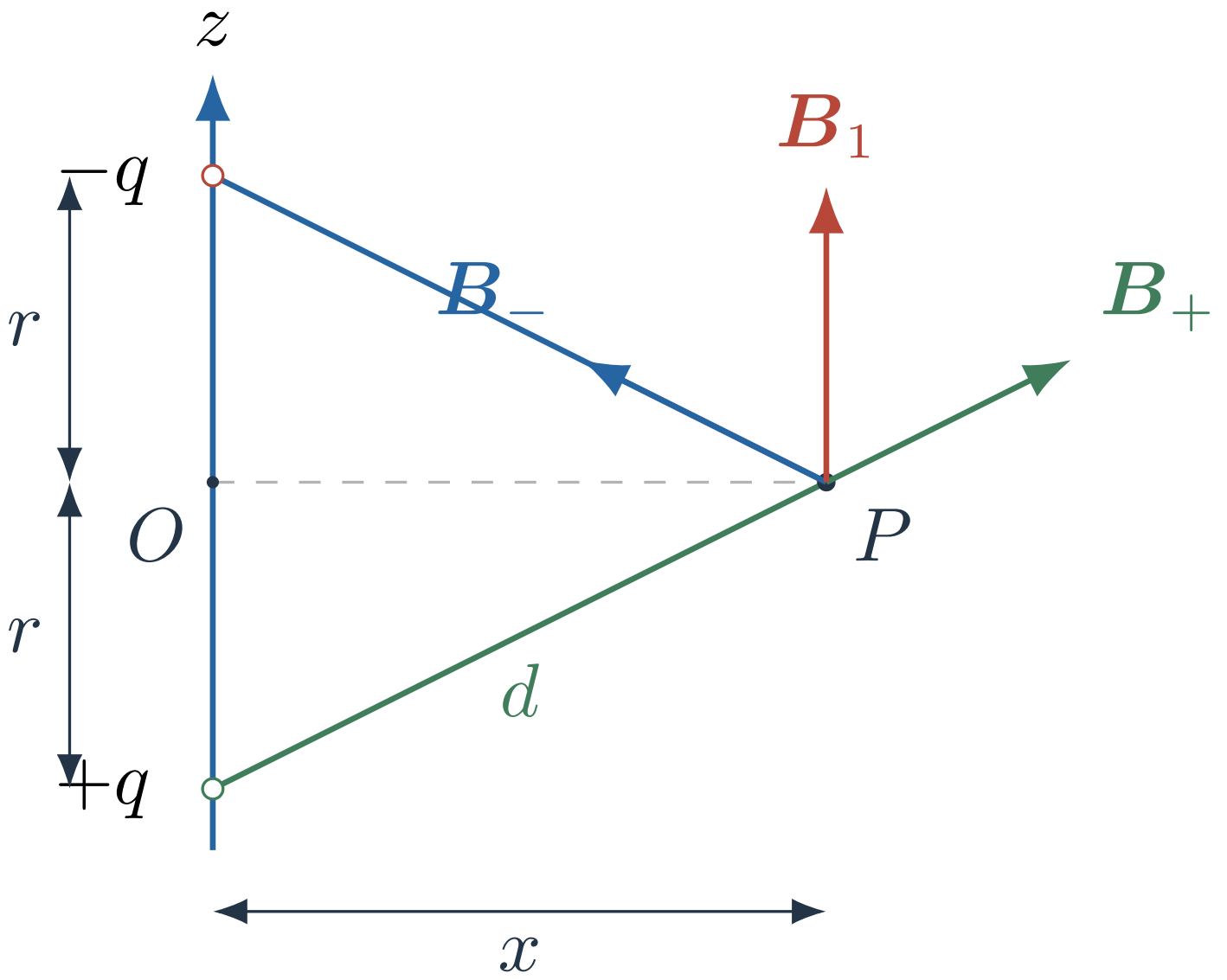
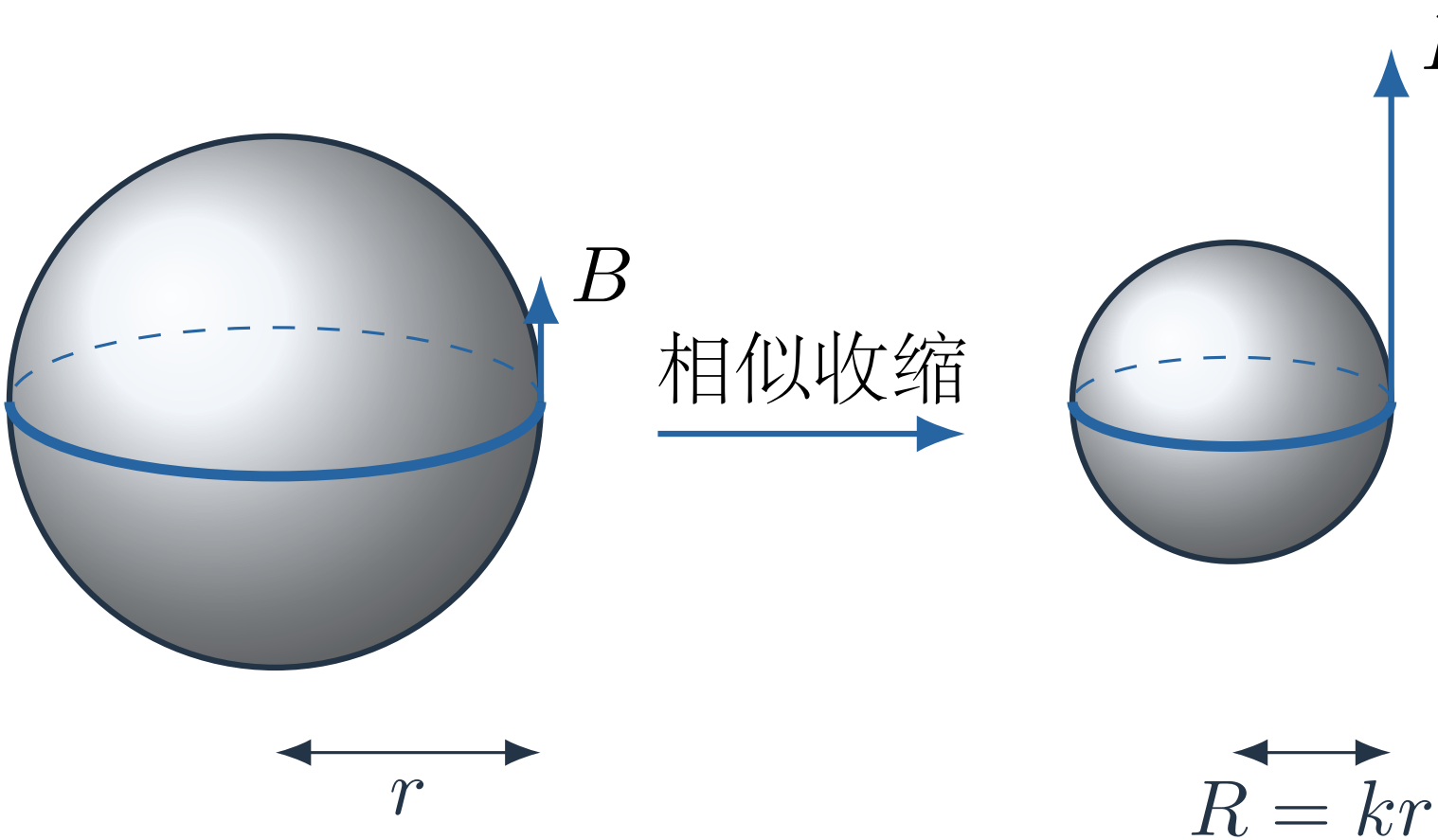


两极为等效源，环与恒星同缩

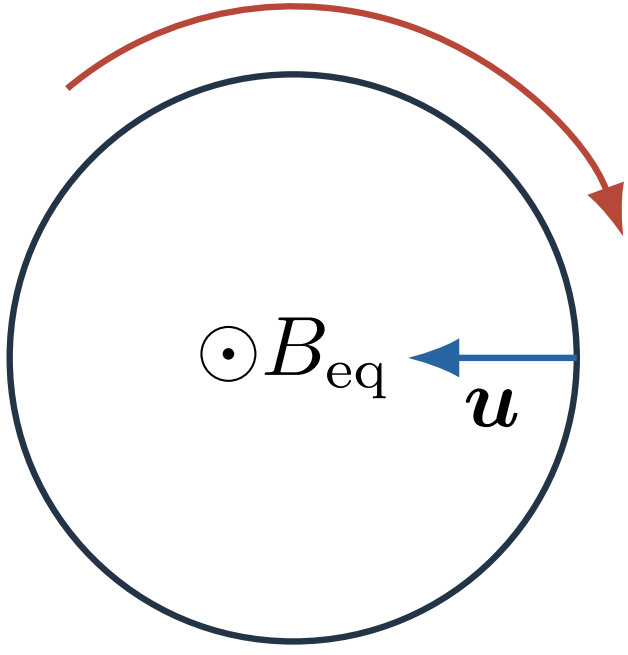

z
−q
r
O
r
+q
B1
B−
B+
P
d
x



面积尺度 $\times k^2$

$B_2R^2 = Br^2$，随动磁通不变

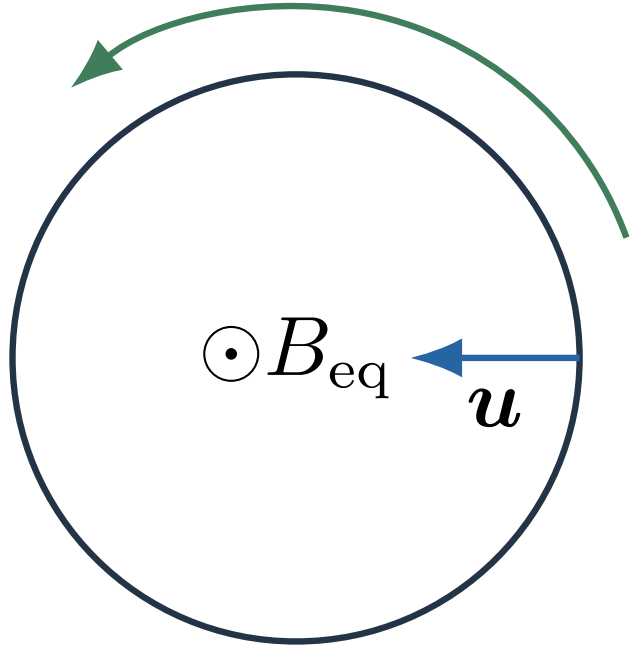
感生项



$\mathcal{E}_{\mathrm{lab}} = 2\pi RaB_{\mathrm{eq}} < 0$

顺时针

运动项



$\mathcal{E}_{\mathrm{mot}} = -2\pi RaB_{\mathrm{eq}} > 0$

逆时针

从 $+z$ 方向看；$a < 0$，总电动势为零